



Ram Racing | Aerodynamic Subteam

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Ansys Fluent Procedure

How to successfully run a simulation in Ansys Fluent 2024R2

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Forward:

This document's goal is to provide the future of Ram Racing with the ability to quickly and accurately use CFD to validate designs. If you have questions that cannot be answered by this document please go to the Ansys customer portal, and find the current Ansys Fluent user guide for a more comprehensive delve into CFD.

Running a Simulation:

Geometry:

Pre-Discovery Steps:

Over the past year, we have found that the most important part of running a successful CFD simulation is the geometry. A separate CFD Assembly should be created with the aerodynamic devices, a simplified solid body chassis, simplified wheels, simplified control arms, and a simplified driver. See *Figure 1* or the CFD Assembly for Dauntless in T:\RamRacing\Design\24-25\Design\IC Subsystems\Aero\CFD\Full Car CFD\Version_of_CFD_car_(updated_chassis)\Full_CarCFD_Assembly.SLDASM for an example.

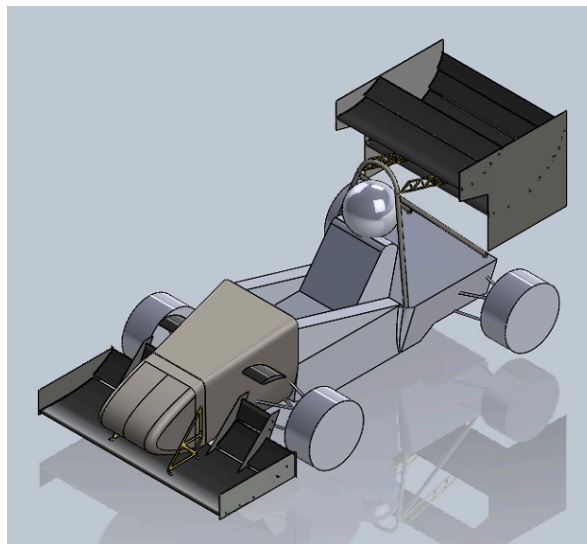


Figure 1, FullCarCFD_Assembly.SLDASM | Dauntless CFD Assembly

Make sure that the Nose Cone, Chassis and Wheels are solid bodies. As in the Nose Cone may be a thin piece of carbon fiber, but the whole thing is a solid block. (See *Figure 2*) Once this assembly has been created with mostly simplified components, you will continue simplification

inside Ansys Discovery, export the assembly as a A214 STEP file and save in a well organized location.

Geometry Work Inside Discovery:

Ansys Discovery is a CAD program that we use for the final simplification of the geometry of the CFD model. There will be three subsections that describe how to create a Full Car Geometry, a Half Car Geometry, and a Quarter Car Geometry. Some of these numbers in these examples will differ depending on the length and height of the car, but all equations that lead us to these numbers will be provided.

Full Car Geometry:

Okay, this is the big one. There are a lot of places where a small inconsistency in your geometry could fuck up your simulation. Best advice we have is take it slow and don't skip any damn steps. This geometry can be used at any speed you wish so once you have a good geometry, use it, iterate off of it, and validate your design.

Step 1: Open the Program...

Open up Ansys Discovery.

Step 2: Import the STEP file

Go to the top left corner of discovery. Click the three lines. Find Import Geometry in the dropdown menu. Locate in the file manager your desired step file and import into the discovery workspace.

Step 3: Change the Models Orientation

This one is really important and stumped us for a while... Because of how Ansys Fluent calculates the projected surface area you must face the front of the car toward the -x, -y, or -z direction. Personally we prefer facing the car toward the -x direction by rotating it 270° (this is because the car in solidworks faces the positive z direction) and making the up direction +y and the left and right directions the z axis. Make sure that you change the center of rotation to the origin of the discovery file. While discovery is smart enough to align the origins of itself and solidworks, it doesn't put the move handle at that origin by default. This orientation has allowed us to go from getting one successful simulation every 2 weeks, to running 5 different simulations in one week.

Step 4: Finalize Simplification

While the Solidworks Assembly you created is pretty simple, it is much easier to finalize the simplification in Discovery since it was built for the purpose of being a CFD centered CAD program.

So, the first step in Discovery Simplification is to fillet the wheels. We usually fillet these to around a 0.5" radii. This allows the wheels to not only act more like wheels in the simulation, but also not cause your simulation to not converge. You also need to make wheel blocks, see Figure 3. Also find the origins of the wheels in meters. Trust me just do it now so you don't hate yourself later.

The second step is to fill literally every hole. If you refer to Figure 4 and 5 you can see an example of a fill operation. Depending on how your parts are designed you might have different parts that need to be filled than the example. Some important ones to not forget are: The remaining chassis tubes (mainly the roll hoop and other tubes that could affect your rear wing), the suspension components, and the wing elements.

Third, simplify all the tabs for the aerodynamic devices. Just make a filleted rectangular prism. Don't forget that simplicity is the goal. Or even better, FUCK THE TABS. Ya dont need em.

Fourth, merge all the bodies. Having one solid body is better in the long run for Ansys, but a good tip is to save the unmerged version just in case you have to make changes.

Fifth, go to the repair tab and go through the fix operations to ensure that there are no errors within the body before we create an enclosure around it.

Overall, you want one body with hopefully zero errors or crazy sharp edges. The more complex you get, the longer it's going to take to mesh, and the worse the mesh will be.

Step 5: Enclosure & Named Selections

Step 6: Export as PMDB File

Inside the Prepare portion of Discovery there should be an export drop down menu. Select Export as PMDB and name the file according to what kind of geometry it is. Store the file in a well organized location.

Computer Settings:

ThreadRipper:

1. Processor Cores: 40 - 50
2. Precision: Double Precision

3. MPI Type: openmpi

Xeon Gold:

1. Processes: 60
2. Precision: Double Precision
3. MPI Type: intel

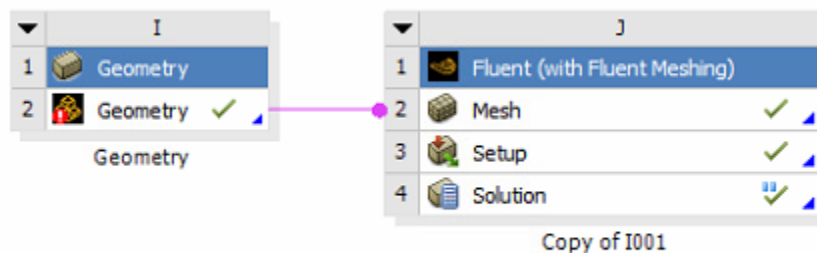
Big Boi:

1. Processes: 128 - 170
2. Precision: Double Precision
3. MPI Type: default

Meshing:

Step 1: Open Meshing Software

Open the Meshing Section of Fluent.



Step 2: Import Geometry

Step 3: Local Refinement

1. Right click Import Geometry and create the new task "Create Local Refinement Regions"

FULL CAR EXAMPLE!!!

If you have half car $z - \min = 0$

Near:

Box

Directly Specify Coords.

Mesh Size [m]: 0.032

X - min	$-(L)$	X - max	$3L$
Y - min	0	Y - max	$H + L/3$
Z - min	$-(W + H/2)$	Z - max	$W + H/2$

Table 1, Near Refinement Size

Mid:

Box

Directly Specify Coords.

Mesh Size [m]: 0.064

X - min	$-(1.25L)$	X - max	$5L$
Y - min	0	Y - max	$H + 2L/3$
Z - min	$-(W + H)$	Z - max	$W + H$

Table 2, Mid Refinement Size

Far:

Box

Directly Specify Coords.

Mesh Size [m]: 0.128

X - min	$-(1.5L)$	X - max	$7L$
Y - min	0	Y - max	$2L$
Z - min	$-(W + 3H/2)$	Z - max	$W + 3H/2$

Table 3, FarRefinement Size

Wheels (If Needed): (DO EACH WHEEL SEPARATELY EX: RLW, RRW, FLW, FRW all with separate refinement zones)

Box

Relative to Body Size:

Mesh Size [m]: 0.032

X - min	0.1	X - max	1
---------	-------	---------	-----

Y - min	0	Y - max	0.1
Z - min	0.1	Z - max	0.1

Table 4, Wheel Refinement Size

Key:

L - Length (x axis)

W - Width (z axis)

H - Height (y axis)

EX:

Create Local Refinement Regions ?

Name: local-refinement-nearfield

Type: Box ▼

Coordinate Specification Method ? : Directly specify coordinates ▼

Mesh Size [mm]: 32

Box Parameters

X Min [mm]	0	X Max [mm]	1500
Y Min [mm]	10.2	Y Max [mm]	2000
Z Min [mm]	-2800	Z Max [mm]	2000

Step 4: Local Sizing

1. Curvature of Stuff:

Growth Rate: 1.2

Size Control Type: Curvature

Local Min Size [m]: 0.001

Max Size [m]: 0.064

Curvature Normal Angle [deg]: 12

Scope To: Faces and Edges

Select By: Label

then select anything that isn't wheels or aero

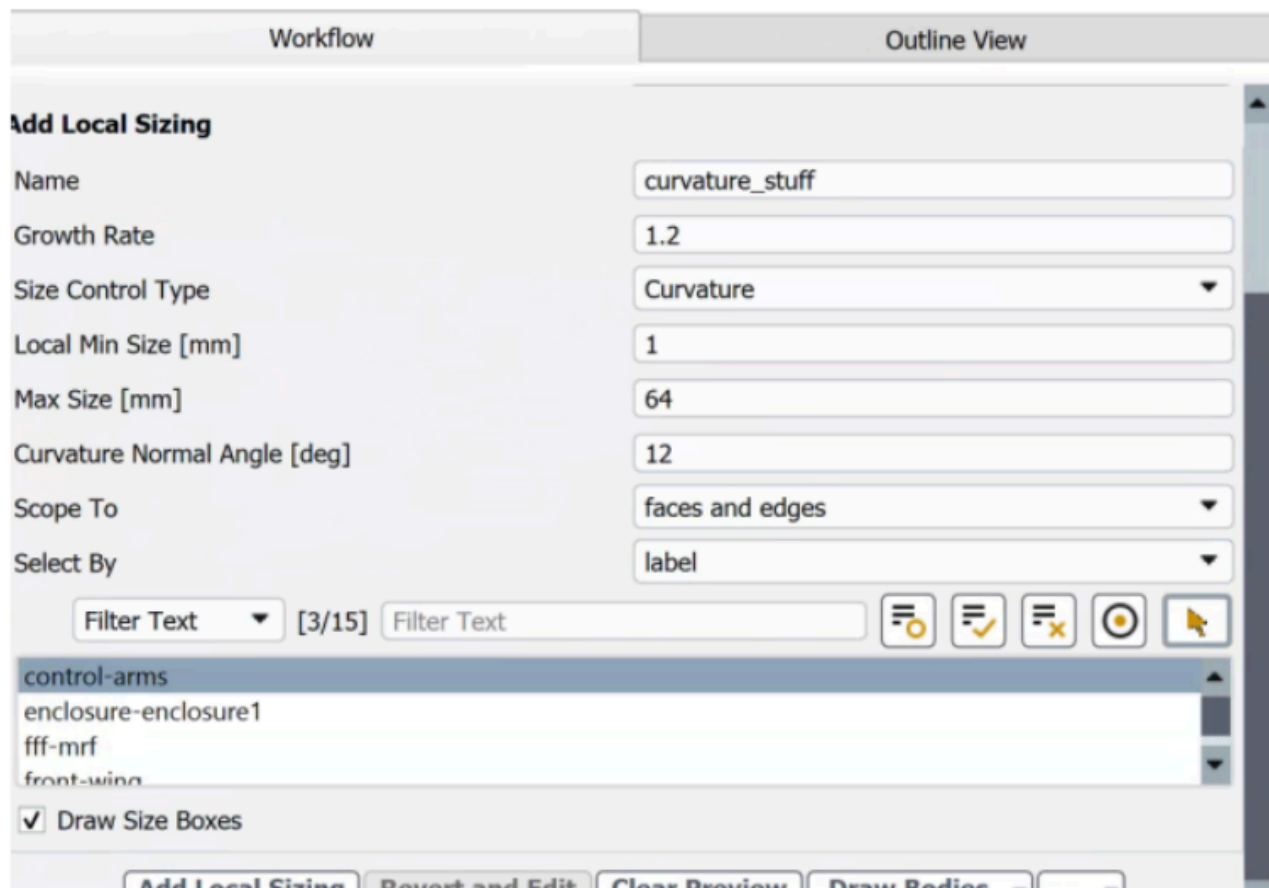


Figure x, Curvature of Stuff

2. Curvature of Aero:

Growth Rate: 1.2

Size Control Type: Curvature

Local Min Size [m]: 0.0005

Max Size [m]: 0.008

Curvature Normal Angle [deg]: 9

Scope To: Faces and Edges

Select By: Label

then select all the aero

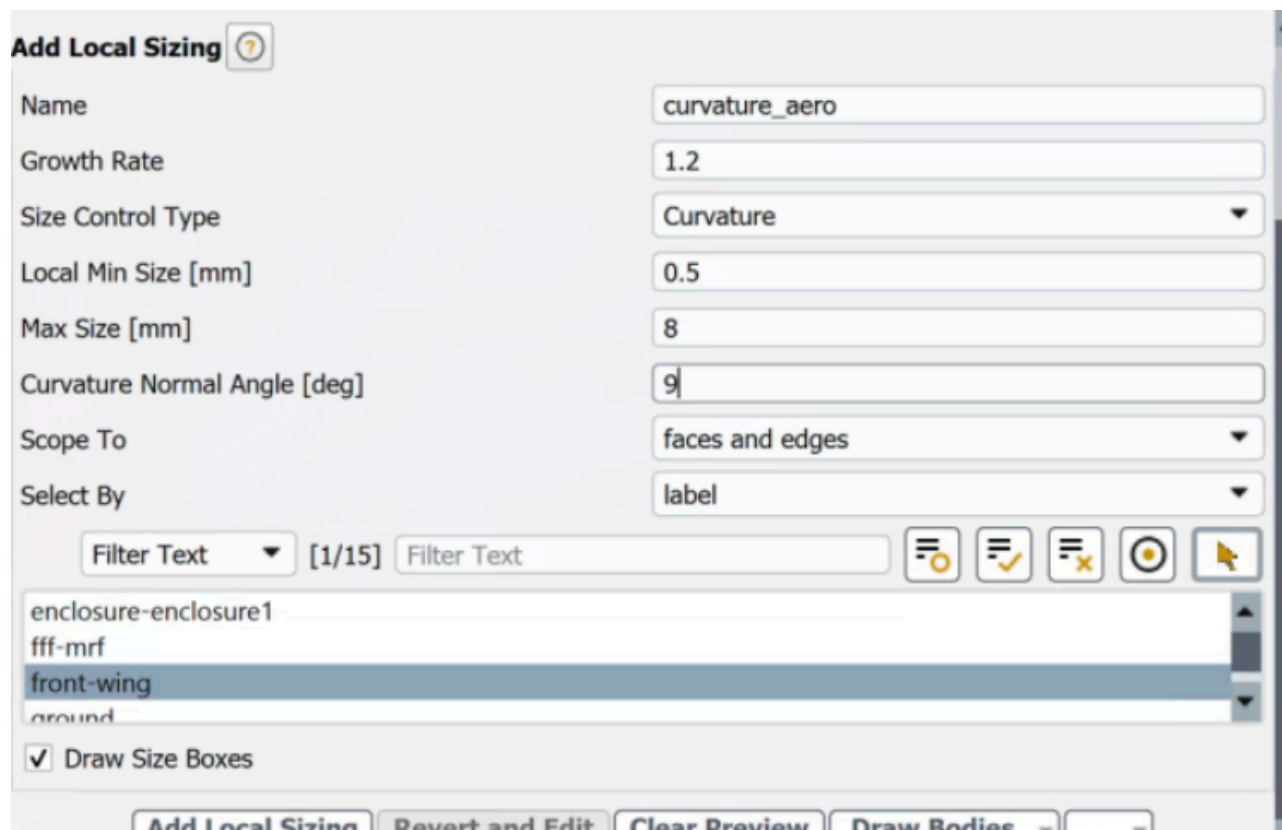


Figure x, Curvature of Aero

3. Curvature of Wheels:

Growth Rate: 1.2

Size Control Type: Curvature

Local Min Size [m]: 0.0005

Max Size [m]: 0.032

Curvature Normal Angle [deg]: 18

Scope To: Faces

Select By: Label

then select da wheel stuff

Add Local Sizing

Name: curvature_wheelsandmrf

Growth Rate: 1.2

Size Control Type: Curvature

Local Min Size [mm]: 0.5

Max Size [mm]: 32

Curvature Normal Angle [deg]: 18

Scope To: faces

Select By: label

Filter Text [2/15] Filter Text

symmetry

top-wall

wheel

☒ Draw Size Boxes

Add Local Sizing Revert and Edit Clear Preview Draw Bodies ...

Figure x, Curvature of Wheels

Step 5: Generate Surface Mesh

Generate the Surface Mesh (?)

Minimum Size [m]: 0.002

Maximum Size [m]: 0.256

Growth Rate: 1.19999

Size Functions: Curvature & Proximity

Curvature Normal Angle [deg]: 18

Cells Per Gap: 1

Scope Proximity To: faces-and-edges

☒ Draw Size Boxes

Separate out Boundary Zones by Angle? No

+ Advanced Options

Figure x, Generate Surface Mesh

Step 6: Improve Surface Mesh

Improve Surface Mesh 

Face Quality Limit

 Advanced Options 

Figure x,

Step 7: Describe Geometry

Geometry Type

☐ The geometry consists of only solid regions

☒ The geometry consists of only fluid regions with no voids

☐ The geometry consists of both fluid and solid regions and/or voids

Change all fluid-fluid boundary types from 'wall' to 'internal'?

☐ Yes

☒ No

Do you want to apply Share Topology?

☐ Yes

☒ No

Enable Multizone Meshing?

☐ Yes

☒ No

Figure x,

Step 8: Update Boundaries

Update Boundaries

Selection Type

label

Filter

Filter Text

Boundary Label	Boundary Type
chassis	wall
frontwing	wall
fw	wall
fwb	wall
ground	wall
inlet	velocity-inlet
outlet	pressure-outlet
rearwing	wall
rw	wall
rwb	wall
symmetry	symmetry
undertray	wall
walls	wall

Figure x,

Step 9: Update Regions

Update Regions

Filter

Filter Text

Region Name	Region Type
enclosure-enclosure1	fluid

Print Summary

Figure x,

IMPORTANT!!!!

If there is more than one region, set the one that is not the enclosure to a solid and ensure that during the generation of the volume mesh that you disable the Generate Solid Regions option.

Step 10: Adding Boundary Layers

Add Boundary Layers ?

Add Boundary Layers? yes

Name last-ratio_1

Offset Method Type last-ratio

Number of Layers 6

Transition Ratio 0.272

First Height [m] 0.0005

Add in fluid-regions

Grow on selected-zones

Zones Filter Text [4/13] Filter Text

- rw
- rwb
- symmetry
- undertrav

Figure x,

Select all the aerodynamic devices and the ground.

Step 11: Generate Volume Mesh

Generate the Volume Mesh ?

Solver Fluent

Fill With poly-hexcore

Peel Layers 1

Min Cell Length [m] 0.0005

Max Cell Length [m] 0.256

☒ Enable Parallel Meshing

☐ Advanced Options

☐ Global Boundary Layer Settings

Figure x,

Step 12: Improve Volume Mesh

Improve Volume Mesh ?

Quality Method: Orthogonal

Cell Quality Limit: 0.2

Do you need additional quality criteria?: No

+ Advanced Options

Figure x,

Setup:

Initial Run:

Step 1: Units & Check Mesh

1. Click the Big Old Check button once it loads
 - a. Press the Check Mesh button

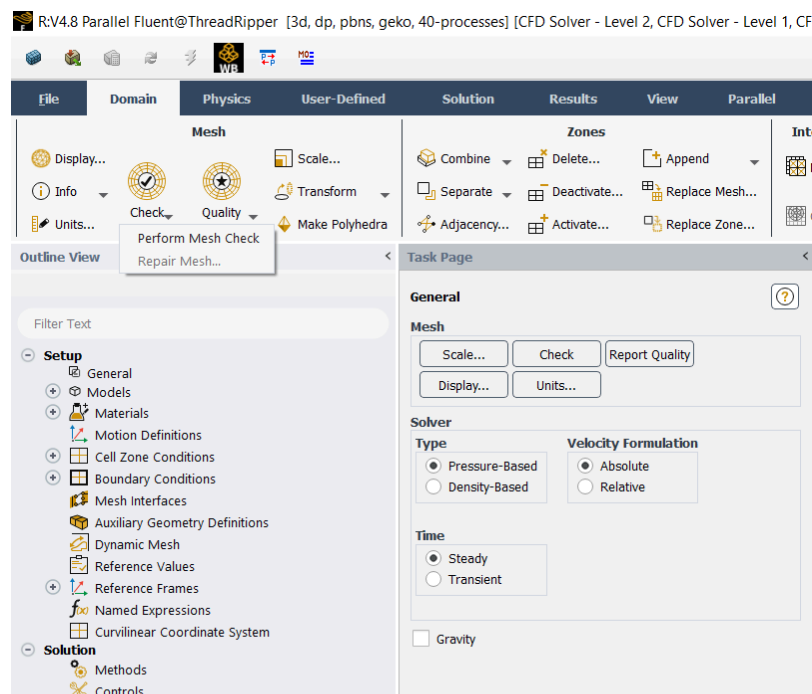


Figure x, Check Mesh button

2. Select the Units Button

3. Find Force and Change the Unit to lbf

Reminder that Half Car Sims do not double the downforce for you so double the result afterward or create a custom unit with a factor of 2.2241

4. Find Velocity and Change the Unit to MpH

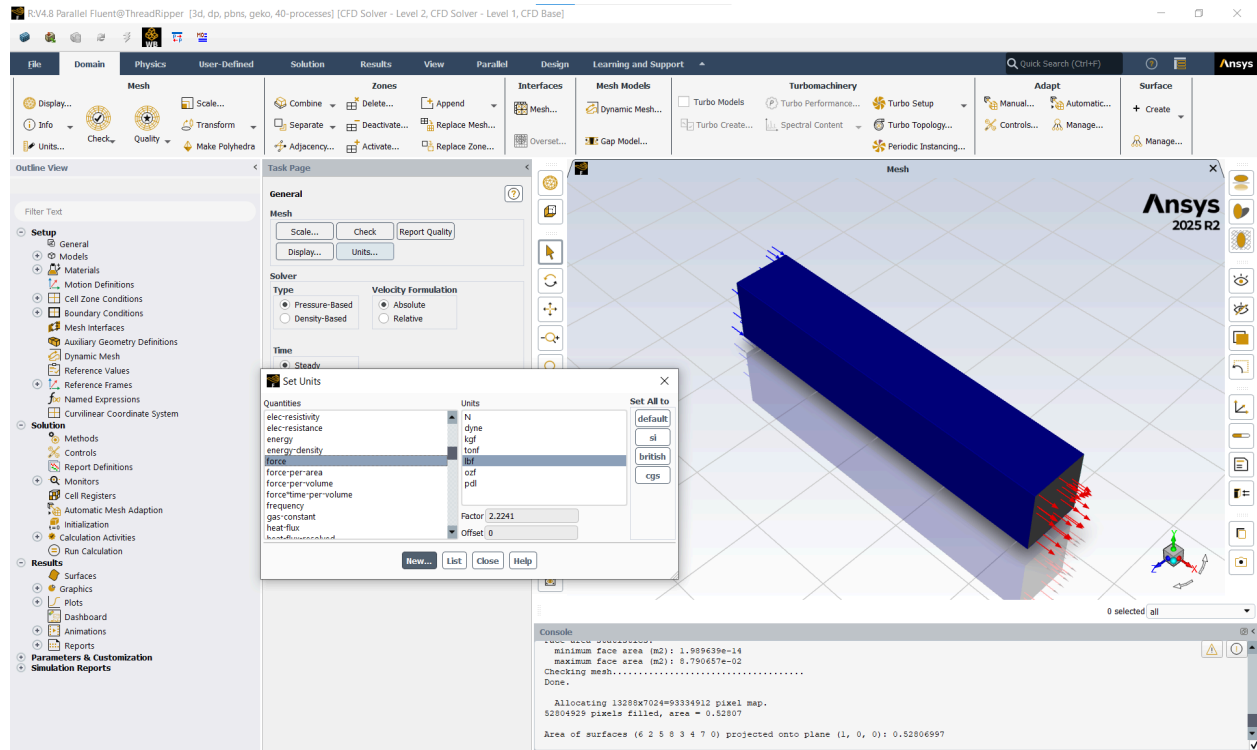


Figure x,

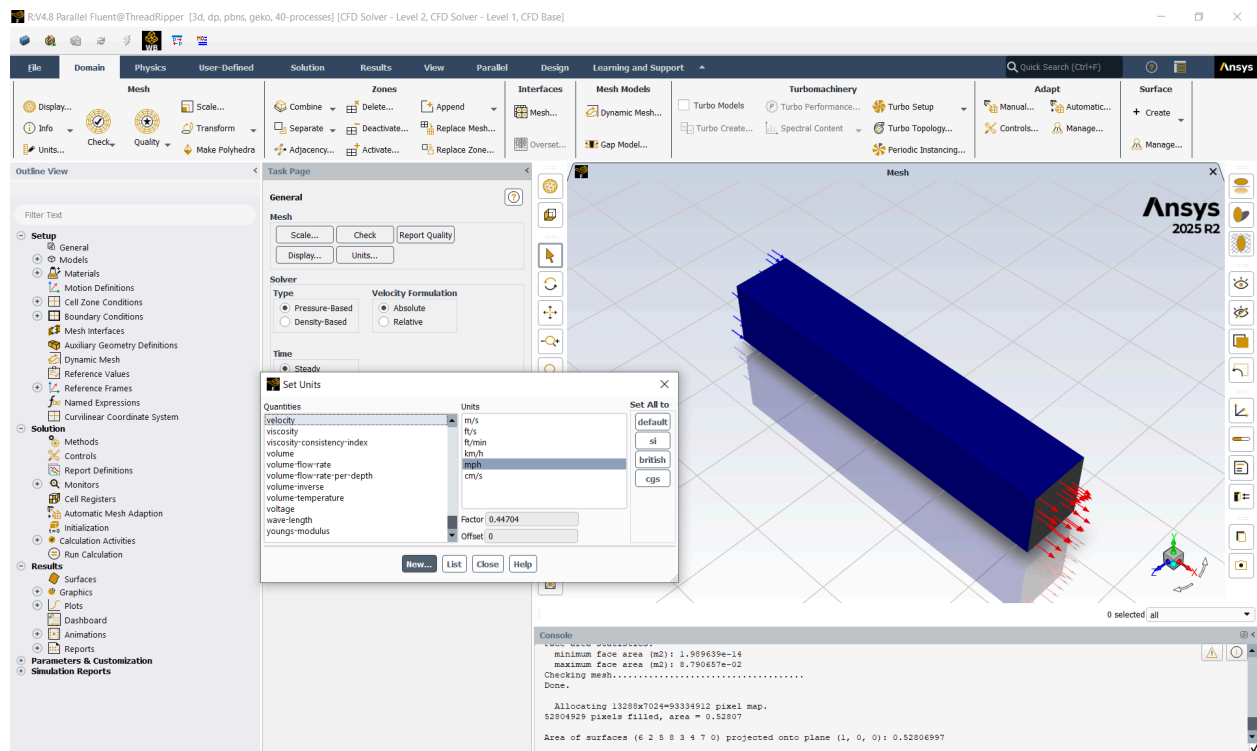


Figure x,

Step 2: Physics Tab

1. Click Reference Values Button

- Set "Compute From" to Inlet
- Set Velocity to 40 mph
- Find total length [m] in SolidWorks and put that value into the Length Section
- Set "Reference Zone" to the enclosure
- Go to the Results Tab
 - Click the Projected Area Button
 - Set the Minimum Feature Size to 0.0001 [m]
 - Select the X Direction
 - Select Chassis, Wheels, Wheel Blocks, Aerodynamic Devices, and all other vehicle zones
 - Press Compute
 - Copy the frontal area number
- Paste the Frontal Area # you just got into the Area Section within Reference Values

2. Return to the Physics Tab
3. Click the Viscous Model Button
 - a. Set Model to k-omega
 - b. Set k-omega Model to GEKO
 - c. Ensure Production Limiter is turned ON
 - d. Ensure Curvature Correction is turned OFF (for now)
 - e. Press OK to save the changes
4. Select the Boundaries Button
 - a. Sort By Zone Type (Top Right Button that looks like an F)
 - b. Select Inlet and Press EDIT
 - i. Set Velocity Magnitude to 40 mph
 - ii. Press Apply
 - c. Select Ground and Press EDIT
 - i. Set to Moving Wall
 - ii. Relative to Adjacent Cell Zone
 - iii. Translational
 - iv. X = 1 Direction
 - v. 40 mph Speed
 - vi. Press Apply
 - d. Wheels (All will be the same between wheels except for the Rotation-Axis Origin)
 - i. Moving Wall
 - ii. Absolute
 - iii. Rotational
 - iv. Speed:

Step 3: Solution Tab

Step 4: Initialize

Step 5: Run

Ramp Up 1 (Second Order w/ Presto Pressure):

Step 1: Physics Tab

Step 2: Solution Tab

Step 3: Run

Ramp Up 2 (Full Second Order w/o Curvature Correction):

Step 1: Physics Tab

Step 2: Solution Tab

Step 3: Run

Ramp Up 3 (Full Send):

Step 1: Physics Tab

Step 2: Solution Tab

Step 3: Run

Solution:

Data You Should Collect:

Downforce & Drag:

Center of Pressure & Aero-Balance:

Yaw, Pitch, & Roll Data:

Per Element Data:

Ride Height Sensitivity:

Aero MAP:

How to Collect Data:

Downforce & Drag:

Center of Pressure & Aero-Balance:

Yaw, Pitch, & Roll Data:

Per Element Data:

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